

Truly Innovative **TECH**

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

Dual-emission laser for automotive and consumer applications

Kyocera SLD Laser has launched a dual-emission white light and infrared (IR) LaserLight source for automotive and consumer lighting, night vision illumination, precision long-range sensing beyond 250 metres and high-speed light fidelity (LiFi) communications



faster than 20Gbps. By delivering both white and IR from the same source and fusing lighting, sensing and communication functionalities, the LaserLight LiFi kit enables intelligent illumination systems that provide accurate sensing and ultra-high-speed communication with unconstrained optical bandwidth, as well as secure and efficient data transmission without RF interference of the Wi-Fi technology. Additionally, dual emission sources can also be used for consumer and professional portable lighting products, night vision illuminators and rangefinders for recreation and outdoor, search and rescue, and security applications.

www.kyocera-sldlaser.com

AI enabled sensor that can sense wide range of gases

Harmful gases and even airborne viruses that negatively impact our health and well-being are present in our modern environment, making air quality poorer. To address this concern, Bosch Sensortec's BME688, an air quality MEMS sensor, combines gas, humidity, temperature, and barometric pressure sensing with innovative artificial intelligence

(AI) capability, creating a small four-in-one air quality measurement solution. Some applications include food spoilage detection, detection of gases emitted due to forest fire, and tracking temperature and humidity changes. The gas sensor can sense the presence of many gases



such as volatile organic compounds, volatile sulphur compounds, and other gas types such as carbon monoxide and hydrogen, in the parts per billion range. Developed for low-power consumption, the single device enables users to reduce their total cost of ownership, cut their development time, and simplify the design.

www.bosch-sensortec.com

Smartphone-operated digital car key module

LG Innotek has developed a digital car key module with improved location detection precision and security that allows wireless data transmission

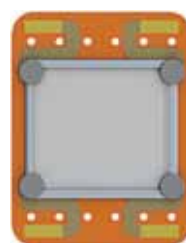


between a car and a smartphone. Users can now oversee several vehicle conditions, including driving distance, fuel efficiency, tyre pressure, etc, at a glance as well as control the vehicle. The digital car key module uses ultra-wideband technology, which is a wireless LAN communication technology, and a proprietary algorithm, improving the location detection precision. Also, the module has increased security by applying proprietary hacking prevention technology.

www.lginnotek.com

Next-gen high-definition piezoelectric film haptic actuator

Enabling design engineers to implement advanced haptic solutions is KEMET's new piezoelectric polymer film haptic actuator which, when embedded



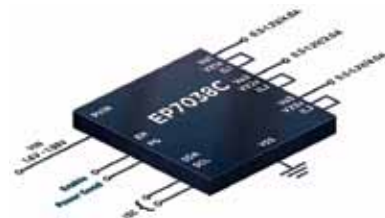
directly into a product's surface, acts as haptic skin that provides localised bodily sensations. The thin (150µm), lightweight, flexible, and conformable haptic actuator

can convey specific material textures and familiar feelings, like the clicks and clacks of buttons, raindrops, and so much more. Unlike eccentric rotating mass technology, which is often used in devices for haptic sensations, the new haptic actuator does not shake the entire device and offers good response time.

<https://ec.kemet.com>

Small integrated voltage regulator with three outputs

Traditional power solutions require dozens of discrete components with big footprints, complex designs and deliver power inefficiently. Empower Semicon-



ductor's high-performance integrated voltage regulator (IVR) EP7038, with three outputs totaling 10A, enables full integration of multiple voltage rails in one IC. The result is 10x space reduction, power delivery with unprecedented simplicity, speed, accuracy, and no discrete